CHAPTER 4:

Parametric Methods



Introduction

- Use the Model is defined with small number of parameters
- Assume some distribution and estimate the parameters of the distribution
- Use the distribution to make decisions

Parametric Estimation

- $\mathcal{X} = \{ x^t \}_t$ where $x^t \sim p(x)$
- Parametric estimation:

  Assume a form for $p(x \mid \theta)$ and estimate θ, its sufficient statistics, using X

  e.g., N (μ, σ^2) where $\theta = \{ \mu, \sigma^2 \}$

Maximum Likelihood Estimation

Bias rule: $p(\theta \mid \mathcal{X}) = p(\mathcal{X} \mid \theta)\, p(\theta) / p(\mathcal{X})$

- ***Likelihood*** of θ given the sample $\mathcal{X}$

$$l(\vartheta \mid \theta) = p(\mathcal{X} \mid \theta) = \prod_t p(x^t \mid \theta)$$

Goal: given $\mathcal{X}$ what is the best θ which describes data

- ***Log likelihood***

$$\mathcal{L}(\theta \mid \mathcal{X}) = \log l(\theta \mid \mathcal{X}) = \sum_t \log p(x^t \mid \theta)$$

- ***Maximum likelihood estimator (MLE)***

$$\theta^* = \text{argmax}_\theta \mathcal{L}(\vartheta \mid \theta)$$

Examples: Bernoulli/Multinomial

- *Bernoulli*: Two states, failure/success, x in {0,1}

  $P(x) = p_o^{\,x} (1 - p_o)^{(1-x)}$

  $$\mathcal{L}(p_o|\mathcal{X}) = \log \prod_t p_o^{\,x^t} (1 - p_o)^{(1-x^t)}$$

  MLE: $p_o = \sum_t x^t / N$

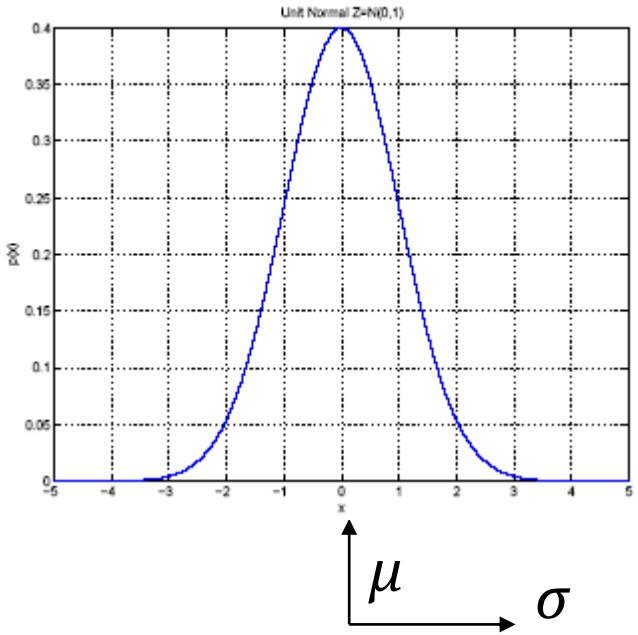
- *Multinomial*: $K>2$ states, x_i in {0,1}

  $P(x_1,x_2,\ldots,x_K) = \prod_i p_i^{x_i}$

  $$\mathcal{L}(p_1,p_2,\ldots,p_K|\mathcal{X}) = \log \prod_t \prod_i p_i^{x_i^t}$$

  MLE: $p_i = \sum_t x_i^t / N$

Gaussian (Normal) Distribution



- $p(x) = \mathcal{N}(\mu, \sigma^2)$

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

- MLE for μ and σ^2:

$$m = \frac{\sum_t x^t}{N}$$

$$s^2 = \frac{\sum_t \left(x^t - m\right)^2}{N}$$

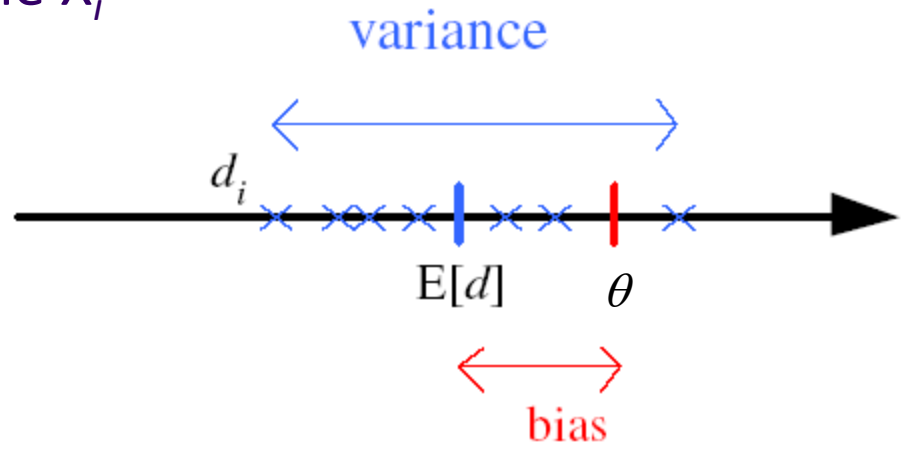
Bias and Variance

Unknown parameter θ

Estimator $d_i = d(X_i)$ on sample X_i

Bias: $b_\theta(d) = E[d] - \theta$

Variance: $E[(d - E[d])^2]$



Mean square error:

$$r(d,\theta) = E[(d-\theta)^2]$$
$$= (E[d] - \theta)^2 + E[(d - E[d])^2]$$
$$= \text{Bias}^2 + \text{Variance}$$

Bayes' Estimator

- Suppose we do know something about the parameter
- Treat θ as a random var with prior $p(\theta)$
- Bayes' rule: $p(\theta|\mathcal{X}) = p(\mathcal{X}|\theta)\, p(\theta) / p(\mathcal{X})$
- Full: $p(x|\mathcal{X}) = \int p(x|\theta)\, p(\theta|\mathcal{X})\, d\theta$
- Maximum a Posteriori (MAP): $\theta_{MAP} = \text{argmax}_\theta\, p(\theta|\mathcal{X})$
- Maximum Likelihood (ML): $\theta_{ML} = \text{argmax}_\theta\, p(\mathcal{X}|\theta)$
- Bayes': $\theta_{\text{Bayes'}} = E[\theta|\mathcal{X}] = \int \theta\, p(\theta|\mathcal{X})\, d\theta$

Bayes' Estimator: Example

- $x^t \sim \mathcal{N}(\theta, \sigma_o^2)$ and $\theta \sim \mathcal{N}(\mu, \sigma^2)$
- $\theta_{ML} = m$
- $\theta_{MAP} = \theta_{Bayes'} =$

$$E[\theta \mid \mathcal{X}] = \frac{N/\sigma_0^2}{N/\sigma_0^2 + 1/\sigma^2} m + \frac{1/\sigma^2}{N/\sigma_0^2 + 1/\sigma^2} \mu$$

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Parametric Classification

$$g_i(x) = p(x \mid C_i) P(C_i)$$

or

$$g_i(x) = \log p(x \mid C_i) + \log P(C_i)$$

$$p(x \mid C_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(x-\mu_i)^2}{2\sigma_i^2} \right]$$

$$g_i(x) = -\frac{1}{2}\log 2\pi - \log \sigma_i - \frac{(x-\mu_i)^2}{2\sigma_i^2} + \log P(C_i)$$

- Given the sample $\mathcal{X}=\{x^t,r^t\}_{t=1}^N$

$$x\in\Re \qquad r_i^t=\begin{cases}1 \text{ if } x^t\in C_i\\ 0 \text{ if } x^t\in C_j, j\neq i\end{cases}$$

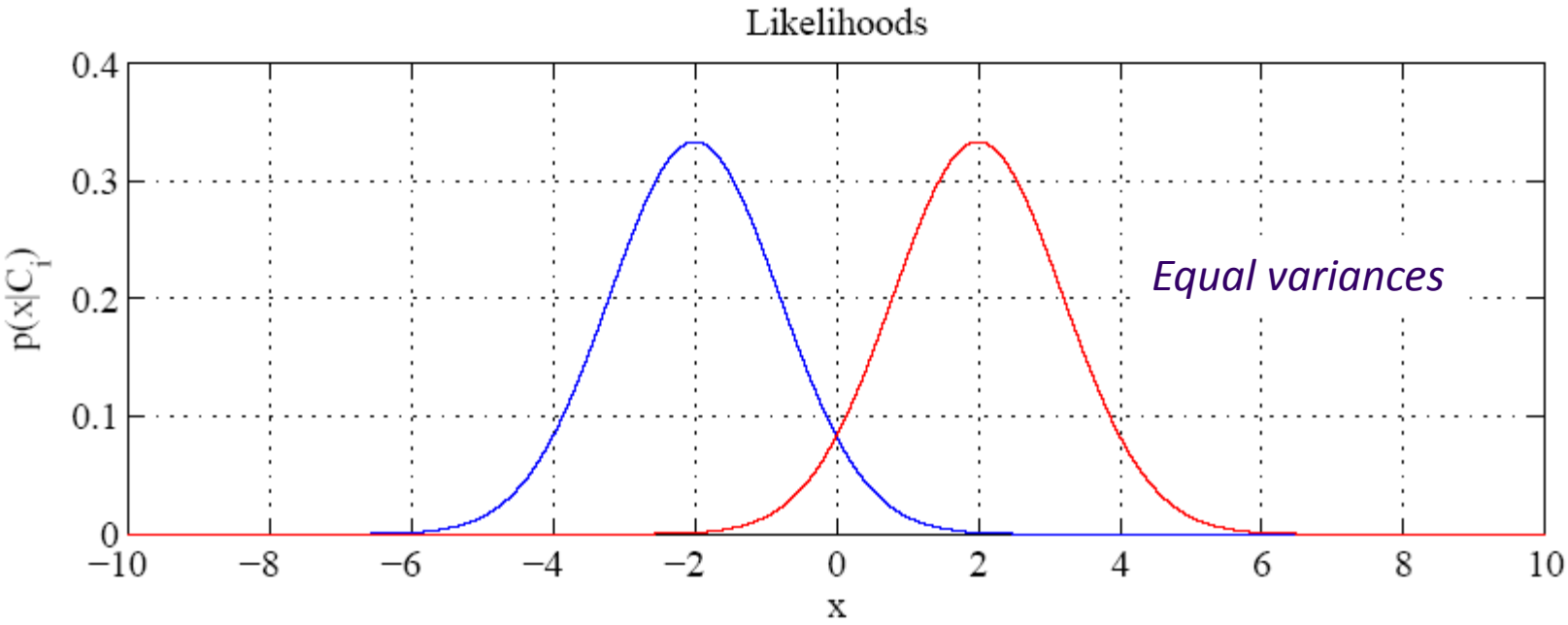
- ML estimates are

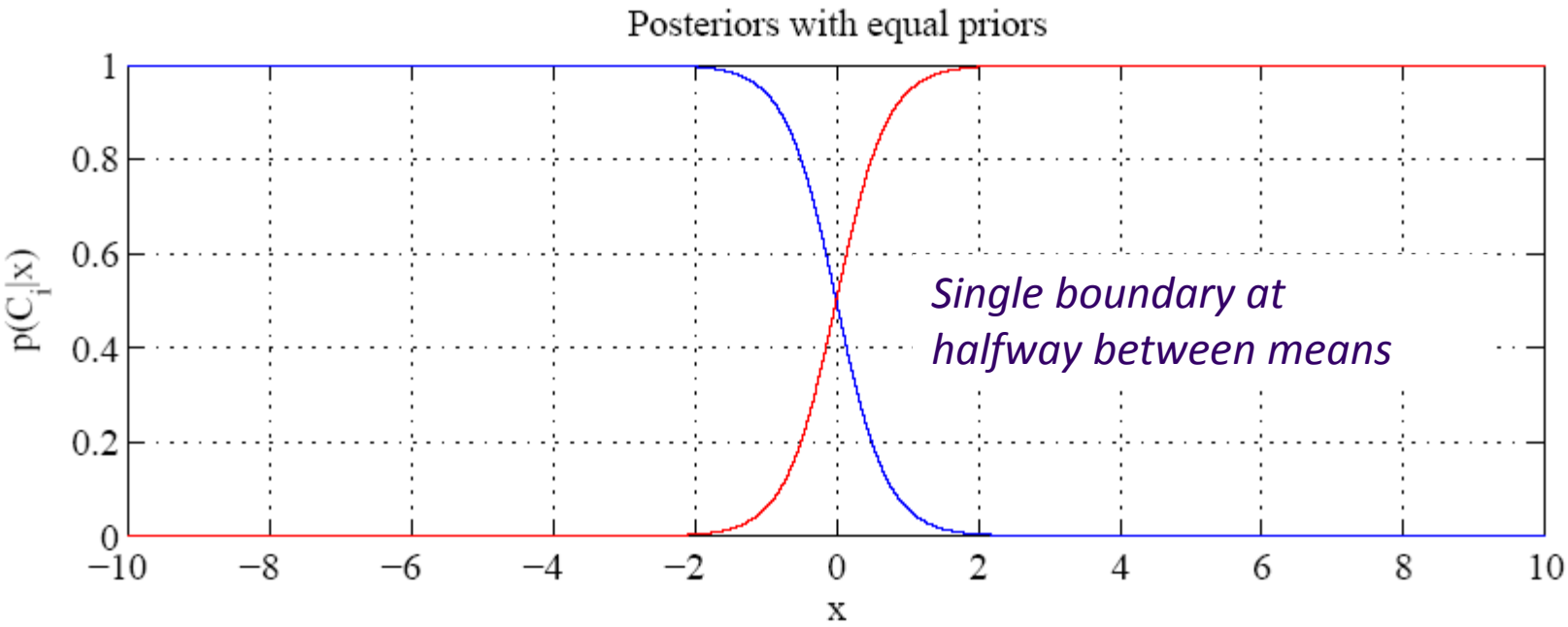
$$\hat{P}(C_i)=\frac{\sum_t r_i^t}{N} \quad m_i=\frac{\sum_t x^t r_i^t}{\sum_t r_i^t} \quad s_i^2=\frac{\sum_t (x^t-m_i)^2 r_i^t}{\sum_t r_i^t}$$

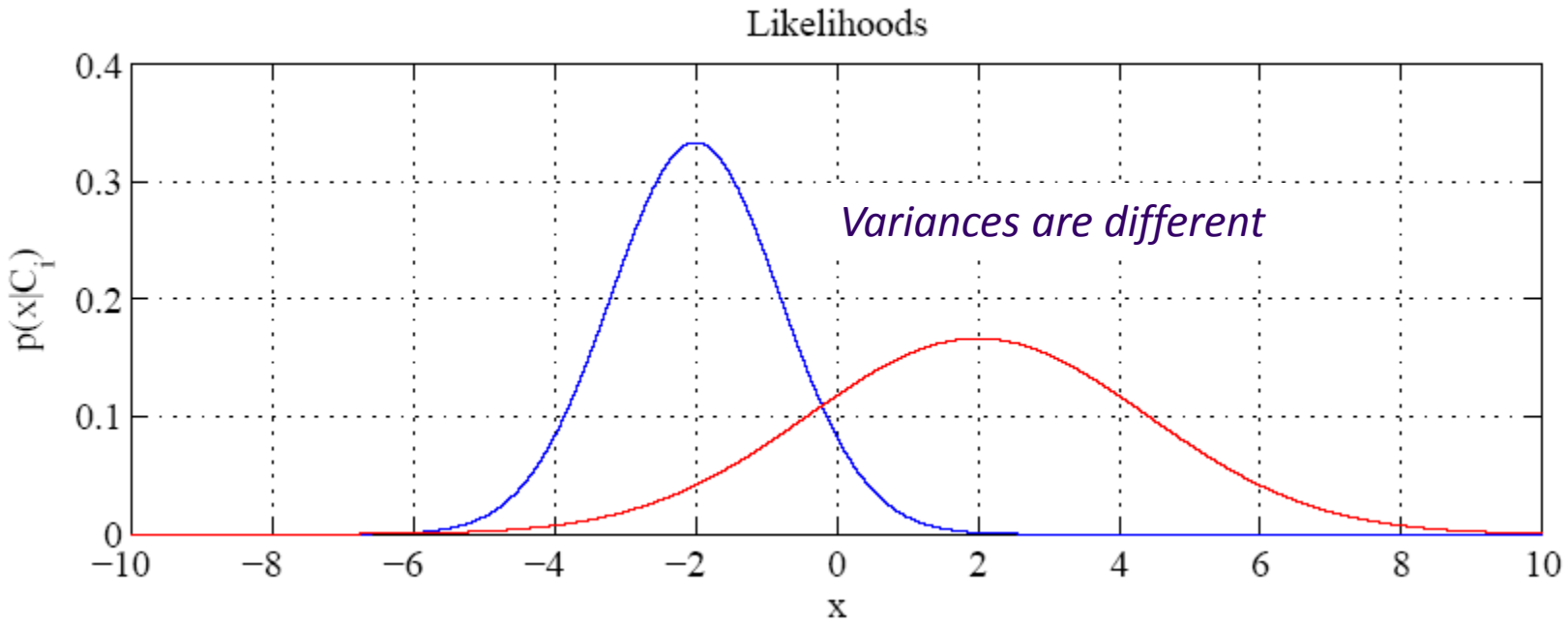
- Discriminant becomes

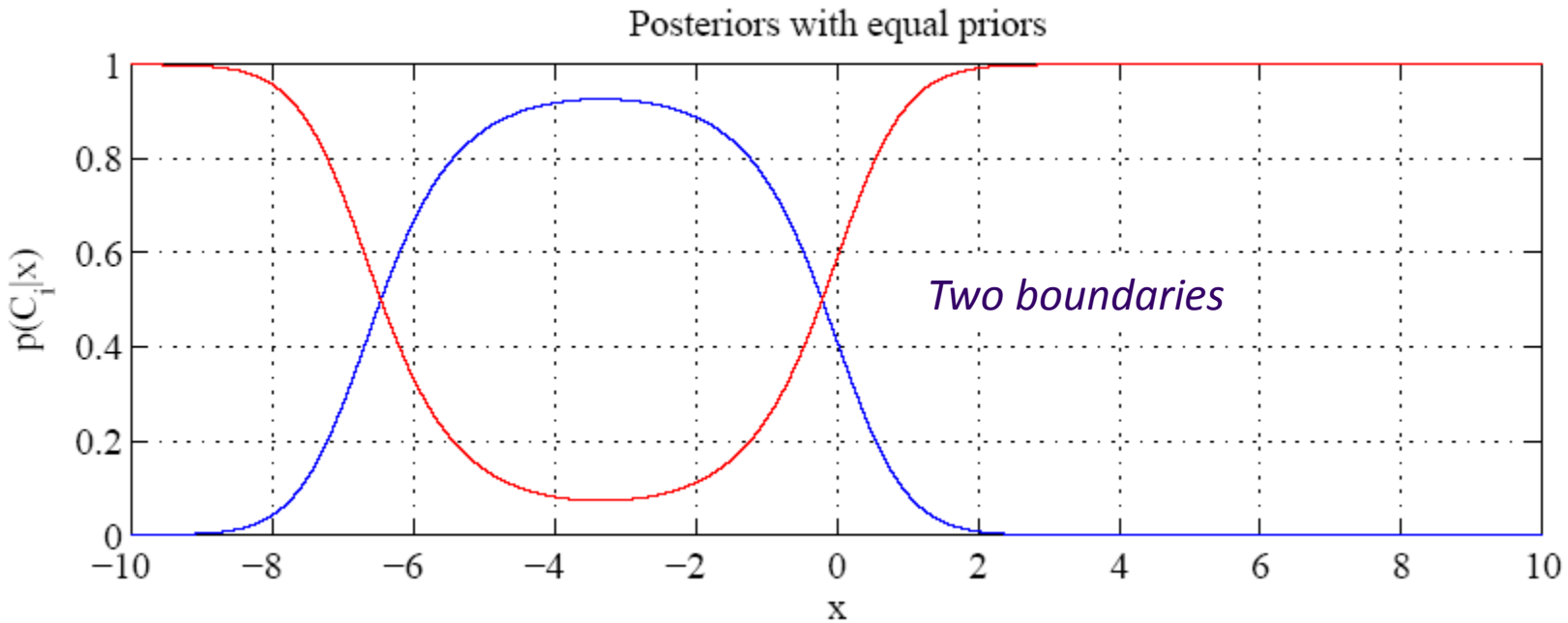
$$g_i(x)=-\frac{1}{2}\log 2\pi-\log s_i-\frac{(x-m_i)^2}{2s_i^2}+\log\hat{P}(C_i)$$

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Likelihoods
0.4
0.3
0.2
0.1
0
p(x|C_i)
−10
−8
−6
−4
−2
0
2
4
6
8
10
x
Equal variances


Posteriors with equal priors
1
0.8
0.6
0.4
0.2
0
p(C_i|x)
−10
−8
−6
−4
−2
0
2
4
6
8
10
x
Single boundary at
halfway between means


Likelihoods
0.4
0.3
0.2
0.1
0
p(x|C_i)
−10
−8
−6
−4
−2
0
2
4
6
8
10
x
Variances are different


Posteriors with equal priors
1
0.8
0.6
0.4
0.2
0
p(C_i|x)
−10
−8
−6
−4
−2
0
2
4
6
8
10
x
Two boundaries